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GRAPH NEURAL NETWORKS FOR AUTONOMOUS THEOREM PROVING

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Contents

Abstract

The main goal of this work is to prove the existence of.....

Notation: We will work over an algebraically closed field K of characteristic zero.

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Chapter 1

Introduction

One of the central challenges in mathematics and computer science is the automatic generation of proofs. This problem has driven the development of proof assistants as Lean, Rocq (formerly Coq) and Agda, systems that allow mathematical theorems to be stated and verified mechanically. In the recent years, due to the increase in capabilities of artificial intelligence models, specially the surge of large language models, there's been a surge in the interest for this problem.

A natural formal setting in which to study this problem is type theory, a foundational framework for mathematics that assigns types to mathematical objects. Type theory was originally introduced to resolve the paradoxes in naive set theory, but it has since evolved into a rich and self-standing foundation for constructive mathematics.

Chapter 2

The Syntax and Semantics of Generalized Algebraic Theories

In this chapter we will introduce Generalized Algebraic Theories, a framework for working with dependent types. They allow us to construct the concept of contextual category, a key base structure for interpreting Calculus of Constructions.

2.1 Generalized Algebraic Theories

Generalized Algebraic Theory (or GAT), originally presented by J. Cartmell in his thesis [1] [2], is an abstraction of Martin-Löf type theory and is a generalization of the usual notion of a many-sorted algebraic theory in the following manner: while in Martin Löff's many-sorted algebraic theories sorts are constant types, interpreted as sets, in generalized algebraic theories sorts might be constant or variable, over a specified range of types, and variable types are interpreted as families of sets. So a generalized algebraic theory consists of:

1. a set of sorts, that can be constant or variable;

2. a set of operators, with the sort of the arguments and value specified;

3. a set of axioms, where each axiom is an identity between similar expressions.

Let's present category theory for example: The set of sorts contains two elements Ob and Hom. Ob is a constant type and Hom is a variable type, for a pair of terms of type Ob, name them x, y, we have a type $Hom(x, y)$. Here we see a key point in general algebraic theories, types vary over terms, no over types. The set of operators also has two elements id and $\circ$. The id operator has one argument x of type Ob, and returns a value of type $Hom(x, x)$. This shows how the type

of value of an operator might vary depending on the arguments. The $\circ$ operator (or composition) expects 5 arguments, three of type Ob, name them x, y, z, one of type $Hom(x, y)$, say f, and one of type $Hom(y, z)$, named g, and it returns a value of type $Hom(x, z)$. Here we see that the type of an argument might also vary depending on a previous argument. Informally, instead of writing $\circ(x, y, z, f, g)$ we can write $\circ(f, g)$, as the first three arguments might be inferred from the last two. This notation is discussed in the original paper [2].

Now we will introduce the notation we will be using when talking about terms and types. This notation will be formally defined in a later section. So consider you have a set of variables V, we will associate terms to variables as required. When asserting that a term t is of type A we will write $t : \Delta$, but if t has variables $x_1, \ldots, x_n$ occurring in it this assertion does not make sense unless we assign some types to $x_1, \ldots, x_m$. To do this we will write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

that reads as: if x_1 is a variable of type Δ_1, ... and if x_n is a variable of type Δ_1, then t is a term of type Δ. Similarly, if we want to assert that if x_1 is a variable of type Δ_1, ... and if x_n is a variable of type Δ_n, then Δ is a type, we can write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

this assertions are called rules or sequents. Rules are used to express which expressions are well-formed. We will work with this rules as units instead of the base expressions. This is because terms and types do not make sense by themselves but they need to be thought of in a context.

Axioms are also written as rules. The conclusion of those rules are equalities between types or between terms of the same type. We write those rules as:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'} \qquad \frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = t' : \Delta}$$

and read as: if x_1 is a variable of type Δ_1, ... and x_n is a variable of type Δ_n, then $\Delta = \Delta'$, or $t = t'$ as terms of type Δ, respectively.

To continue with the previous example, we can write the axioms of category theory as:

$$\begin{aligned}
&x, y : Ob, f : Hom(x, y), \quad \circ(id(x), f) = f \\
&x, y : Ob, f : Hom(x, y), \quad \circ(f, id(y)) = f \\
&w, x, y, z : Ob, f : Hom(w, x), g : Hom(x, y), h : Hom(y, z), \\
&\qquad \circ(\circ(f, g), h) = \circ(f, \circ(g, h))
\end{aligned}$$

When presenting a theory's language i.e. the sets of sorts and operators, we must present each symbol with a rule specifying the role of this symbol, either

as a sort symbol or as specifically typed operator symbol. This rules are called introductory rules. In the case of a sort symbol A the rule should be like

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, x_n) \text{ is a type}}$$

clearly specifying the types A is dependent on. In the case of an operator symbol, a rule of the form

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{f(x_1, \ldots, x_n) : \Delta}$$

suffices to specify the types of the arguments and the value of f.

Finally, to present a theory we will give a set os symbols, each associated to its introductory rule, and a set of axioms. And, of course, everything must be well-formed, something we will address in the following section. So now we can give a formal presentation of the theory of categories:

Symbol	*Introductory Rule*
Ob	Ob is a type.
Hom	$x, y \in \text{Ob} : \text{Hom}(x, y)$ is a type.
$\circ$	$x, y, z \in \text{Ob}$, $f \in \text{Hom}(x, y)$, $g \in \text{Hom}(y, z) : \circ(f, g) \in \text{Hom}(x, z)$.
id	$x \in \text{Ob} : \text{id}(x) \in \text{Hom}(x, x)$.

Axioms

$\circ(\text{id}(x), f) = f$,	whenever $x, y \in \text{Ob}$ and $f \in \text{Hom}(x, y)$.
$\circ(f, \text{id}(y)) = f$,	whenever $x, y \in \text{Ob}$ and $f \in \text{Hom}(x, y)$.
$\circ(\circ(f, g), h) = \circ(f, \circ(g, h))$,	whenever $w, x, y, z \in \text{Ob}$, $f \in \text{Hom}(w, x)$, $g \in \text{Hom}(x, y)$ and $h \in \text{Hom}(y, z)$.

2.2 Formal Definition

In this section we will formally define all of the concepts we just introduced. To give a notion of well-formed introductory rule, we need the concept of drivability, which also depends on the introductory rules. To tackle this problem we will leave aside the question of well-formedness until we have the complete set of derived rules of a theory. We will first define a pretheory and then define a theory as a well-formed pretheory.

We will assume to have countable set of variables V. Lets define the notion of rule on an alphabet W, the elements of W are called symbols.

Definition 2.1. *The set of **expressions** is defined inductively such that expressions are finite sequences of elements of $W \cup V \cup \{(\} \cup \{)\} \cup \{,\}$ defined by the clauses:*

1. *if $x \in V$, then x is an expression;*
2. *if $L \in W$, then L is an expression;*
3. *if $L \in W$ and $e_1, \ldots, e_n$ are expressions, then $L(e_1, \ldots, e_n)$ is an expression.*

Now we can define the building blocks of a rule:

Definition 2.2. *A **premise** is defined to be a sequence of $V\times$ the set of expressions. The empty sequence is allowed and we call it an empty premise. If $((x_1, \Delta_1), \ldots, (x_n, \Delta_n))$ is a premise, we write it as $x_1 : \Delta_1, \ldots, x_n : \Delta_n$.*

Definition 2.3. *We call a **conclusion** to one of the following:*

1. *T-conclusion: defined by a single expression Δ and written as 'Δ is a type'.*
2. *$\in$-conclusion: defined by a pair of expressions (t, Δ) and written as '$t \in \Delta$'.*
3. *T =conclusion: defined by a pair of expressions (Δ_1, Δ_2) and written as '$\Delta_1 = \Delta_2$'.*
4. *$\in$=conclusion: defined by a triple of expressions (t_1, t_2, Δ) and written as '$t_1 = t_2 : \Delta$'*

Finally a **rule** is determined by a premise P and a conclusion C, and is written as $\dfrac{P}{C}$. We name the rule as T-rule, $\in$-rule, T =rule or $\in$=rule according to the form of the conclusion.

Now we can define pretheory and later theory.

Definition 2.4. *A **pretheory** consists of*

1. *a set of sort symbols S;*
2. *a set of operator symbols Z;*
3. *for each sort symbol A, an associated rule of the alphabet $S \cup Z$:*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, A_n) \text{ is a type}}$$

for some $n \geq 0$. It is called the introductory rule of A;

4. *for each operator symbol F, an associated rule of the alphabet $S \cup Z$:*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{F(x_1, \ldots, x_n) : \Delta}$$

for some $n \geq 0$. It is called the introductory rule of F;

5. *a set of axioms. Each axiom is either a T =rule or a $\in$=rule of the alphabet $S \cup Z$.*

Now we will tackle the problem of well-formedness of rules.

Definition 2.5. *If U is a pretheory, then*

1. *a **context** is a premise $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ such that the rule*
$$\frac{x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}}{\Delta_n \text{ is a type}}$$
is a derived rule such that x_n is distinct from all of $x_1, \ldots, x_n$.

2. (a) *the rule $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ is **well-formed** if the premise $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ is a context.*
   (b) *the rule $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$ is **well-formed** if $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ is a derived rule of U.*
   (c) *the rule $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}$ is **well-formed** if $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ and $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta' \text{ is a type}}$ are derived rules of U.*
   (d) *the rule $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = t' : \Delta}$ is **well-formed** if $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$ and $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t' : \Delta}$ are derived rules of U.*

Definition 2.6. *The set of **derived rules** of a pretheory U is the set of rules derivable by the following principles of derivation:*

Definition 2.7. *A pretheory is well-formed if all of its introductory rules are well-formed. A **theory** is a well-formed pretheory.*

Lastly, we will introduce substitution. Given $\Delta, t_1, \ldots, t_n$ expressions and $x_1, \ldots, x_n$ variables, then we define the expression $\Delta[x_1|t_1, \ldots, x_n|t_n]$ the result of simultaneous replacement of $x_1, \ldots, x_n$ in Δ by $t_1, \ldots, t_n$. We call this the Substitution operation. Then, as proven in the original paper [2], the set of derived rules of a theory is closed under the operation of substitution of correctly typed terms for variables. We could add substitution as one of the derivation rules but would make applying induction very messy.

2.3 Models and Homomorphisms

With all the syntax defined, we will consider the semantics of a theory. To do this, we will introduce the notions of model and model homomorphism.

First, we will set some notation and basic concepts.

Definition 2.8. *A tree is a pair $(N, father)$ where N is a not empty set of nodes and $father : N \to N$ such that:*

1. *for all $A \in N$ there exists $n \in \omega$ such that $father^n(A) = father^{n+1}(A)$;*
2. *for all $A, B \in N$, if $A = father(A)$ and $B = father(B)$, then $A = B$.*

We will write $A \trianglelefteq B$ if $A = father(B)$ and $A \triangleleft B$ if $A \trianglelefteq B$ and $A \neq B$. By condition 2, there is a unique least element of N, call it 1. Finally we will define, for any $A \in N$, $level(A)$ to be the least $n \in \omega$ with $father^n(A) = father^{n+1}(A)$.

The set of contexts of a theory is structured as a tree. Whit the empty context as it's least element and $father(\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle) = \langle x_1 : \Delta_1, \ldots, \Delta_{n-1} \rangle$.

We will also want to consider the tree of sets, families of sets, families of families of sets, ...Consider the following notation for families: given a set A, if for every element $a \in A$ $B(A)$ is a set, then the corresponding A-indexed family of sets is denoted as $\lambda a \in A.B(a)$ or as $\lambda a.B(a)$; If we have a set $C(a, b)$ for every $a \in A$ and $b \in B(a)$, then $\lambda a \in A.\lambda b \in B(a).C(a, b)$ is an A-indexed family of families of sets; and so on. The collection of sets, families of sets, families of families of sets, and so on, is called the **tree of families**. Given $1 \triangleleft A_1 \triangleleft A_2 \triangleleft \cdots \triangleleft A_n \triangleleft A$ in the tree of families. Then, for all $a_1 \in A_1$, ..., for all $a_n \in A_n(a_1, \ldots, a_{n-1})$, $A(a_1, \ldots, a_n)$ is a set. Finally we should define the concept of **operator**. If we have $1 \triangleleft A_1 \triangleleft \cdots \triangleleft A_n \triangleleft A$ in the tree of families. If for all $a_1 \in A_1$, ..., for all $a_n \in A_n(a_1, \ldots, a_{n-1})$, $f(a_1, \ldots, a_n) \in A(a_1, \ldots, a_n)$. We say that $\lambda a_1 \in A_1 \ldots \lambda a_n \in A_n(a_1, \ldots, a_{n-1}).f(a_1, \ldots, a_n)$ is an operator at A.

Now we have the necessary structure to define a model of a theory. It is necessary to state the fact that models do not interpret expressions, they interpret the derived rules of the theory. The interpretation of a rule R within a model M is written as R^M.

Definition 2.9. *Let U be a theory, a **model** M is a partial mapping from the set of derived rules of U to Fam, such that:*

1. *if R is a derived T-rule $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$, if we name R_i the rules $\dfrac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i \text{ is a type}}$ for $1 \leq i \leq n$. Then $1 \triangleleft R_1^M \triangleleft R_2^M \triangleleft \cdots \triangleleft R_n^M \triangleleft R^M$ in the tree of families.*
2. *if R_t is a derived $\in$-rule $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$ the interpretation R_t^M is an operator at R^M.*

2.4 Contexts and Realizations

In order to construct the contextual category of a theory U, we will first define the category of contexts and realizations of U. This category, noted $R(U)$ has as

objects the contexts of U. Given two contexts $\Gamma \equiv x_1 \colon A_1, \ldots, x_n \colon A_n$ and $\Delta \equiv y_1 \colon B_1, \ldots, y_m \colon B_m$ of U, a morphism from Γ to Δ, or a *realization* of Δ from Γ, is a m-tuple $\langle t_1, \ldots, t_n \rangle$ such that for each j, $1 \leq j \leq m$, the rule

$$\vdash \Gamma \Rightarrow t_j \colon B_j[y_1 \mid t_1, \ldots, y_{j-1} \mid t_{j-1}]$$

The identity of an object $x_1 \colon \Delta_1, \ldots, x_n \colon \Delta_n$ is the realization $\langle x_1, \ldots, x_n \rangle$, and composition is defined as

$$\langle t_1, \ldots, t_m \rangle \circ \langle s_1, \ldots, s_w \rangle = \langle s_1[y_1 \mid t_1, \ldots, y_m \mid t_m], \ldots, s_w[y_1 \mid t_1, \ldots, y_m \mid t_m]$$

whenever $\Gamma = x_1 \colon A_1, \ldots, x_n \colon A_n$, $\Delta = y_1 \colon B_1, \ldots, y_m \colon B_m$ and $\Omega = z_1 \colon C_1, \ldots, z_w \colon C_w$ are contexts, and $\langle t_1, \ldots, t_m \rangle : \Gamma \to \Delta$ and $\langle s_1, \ldots, s_w \rangle : \Delta \to \Omega$ are realizations.

This category has two important properties we should emphasize in:

Given the following definition of tree:

Definition 2.10. *A tree is a pair $(N, father)$ where N is a not empty set of nodes and $father : N \to N$ such that:*

1. *for all $A \in N$ there exists $n \in \omega$ such that $father^n(A) = father^{n+1}(A)$;*
2. *for all $A, B \in N$, if $A = father(A)$ and $B = father(B)$, then $A = B$.*

We will write $A \trianglelefteq B$ if $A = father(B)$ and $A \triangleleft B$ if $A \trianglelefteq B$ and $A \neq B$. By condition 2, there is a unique least element of N, call it 1. Finally we will define, for any $A \in N$, $level(A)$ to be the least $n \in \omega$ with $father^n(A) = father^{n+1}(A)$. We can see that the set of objects of $R(U)$, i.e. the set of contexts of U is structured as a tree Whit the empty context as it's least element and $father(\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle) = \langle x_1 : \Delta_1, \ldots, \Delta_{n-1} \rangle$. Moreover, given a context $x_1 \colon A_1, \ldots, x_n \colon A_n, x_{n+1} \colon A_{n+1}, \ldots x_{n+m}$ of U, $\langle x_1, \ldots, x_n \rangle$ is a realization of $x_1 \colon A_1, \ldots, x_n \colon A_n$ with regard to $x_1 \colon A_1, \ldots, x_n \colon A_n, x_{n+1} \colon A_{n+1}, \ldots x$ and is noted as $p(\langle x_1 \colon A_1, \ldots, x_{n+m} \colon A_{n+m} \rangle, \langle x_1 \colon A_1, \ldots, x_n \colon A_n \rangle)$ Hence, given two objects $\Gamma, \Delta \in \mathrm{Ob}R(U)$, such that $\Gamma \leq \Delta$, $p(\Delta, \Gamma)$ is defined and $p(\Delta, \Gamma) : \Delta \to \Gamma$ in $R(U)$.

And if $\Gamma \equiv x_1 \colon A_1, \ldots, x_n \colon A_n$, $\Gamma' \equiv y_1 \colon B_1, \ldots, y_m \colon B_m$ and $\Delta \equiv y_1 \colon B_1, \ldots, y_{m+w} \colon B_{m+w}$ are contexts, and if $f = \langle t_1, \ldots, t_m \rangle : \Gamma \to \Gamma'$ in $R(U)$, then

$$\begin{array}{ccc} f^*\Delta & \xrightarrow{q(f,\Delta)} & \Delta \\ {\scriptstyle p(f^*\Delta,\Gamma)}\downarrow & & \downarrow{\scriptstyle p(\Delta,\Gamma')} \\ \Gamma & \xrightarrow{f} & \Gamma' \end{array}$$

is a pullback diagram in $R(U)$, where

$$f^*\Delta \equiv x_1 \colon A_1, \ldots, x_n \colon A_n, \\ y_{m+1} \colon B[y_1 \mid t_1, \ldots, y_m \mid t_m], \ldots, y_{m+w} \colon B[y_1 \mid t_1, \ldots, y_m \mid t_m]$$

$$q(f, \Delta) = \langle t_1, \ldots, t_m, y_{m+1}, \ldots, y_{m+w} \rangle$$

This motivates the following definition

2.5 Contextual Categories

Definition 2.11. *a **contextual category** consists of*

1. *a category $\mathcal{C}$ with a terminal object 1;*
2. *a function father: $Ob\mathcal{C} \to Ob\mathcal{C}$ such that $(Ob\mathcal{C}, father)$ is a tree, and 1 is the root of the tree, i.e. $father 1 = 1$;*
3. *for each object A of $\mathcal{C}$ different from 1, a morphism $p(A)\colon A \to father A$ called canonical projection of A, we write it as $A \twoheadrightarrow father(A)$;*
4. *for all objects A, B in $\mathcal{C}$ with $A \lhd B$ and all morphisms $f\colon C \to A$, an object f^*B of $\mathcal{C}$ such that $C \lhd f^*B$, and a morphism $q(f, B)\colon f^*B \to B$ in $\mathcal{C}$ such that*

$$\begin{array}{ccc} f^*B & \xrightarrow{q(f,B)} & B \\ \downarrow & & \downarrow \\ C & \xrightarrow{f} & A \end{array}$$

is a pullback in $\mathcal{C}$ that satisfies the following functoriality conditions:

(a) if A and B are objects in $\mathcal{C}$ such that $A \lhd B$, then

$$id_A^* B = B \quad \text{and} \quad q(id_A, B) = id_B$$

(b) whenever

$$\begin{array}{ccccc} & & & & B \\ & & & & \downarrow \\ A & \xrightarrow{f} & A' & \xrightarrow{f'} & A'' \end{array}$$

in $\mathcal{C}$, then

$$(f' \circ f)^* B = f^*(f'^* B) \quad \text{and} \quad q(f' \circ f, B) = q(f', B) \circ q(f, f'^* B)$$

We have that $R(U)$ is structured as a contextual category. The problem with $R(U)$ is that semantically equal but syntactically different rules are interpreted as different elements. Hence we will define an equivalence relation between derived T- and $\in$- rules such that:

$$x_1\colon A_1, \ldots, x_n\colon A_n \Rightarrow A_{n+1}\text{type} \equiv y_1\colon B_1, \ldots, y_m\colon B_m \Rightarrow B_{m+1}\text{type}$$

if, and only if, $m = n$ and for each $1 \leq i \leq n+1$,

$$\vdash x_1 \colon A_1, \ldots, x_{i-1} \colon A_{i-1} \Rightarrow A_i = B_i[y_1 \mid\ x_1, \ldots, y_{i-1} \mid\ x_{i-1}]$$

and

$$x_1 \colon A_1, \ldots, x_n \colon A_n \Rightarrow t : A \equiv y_1 \colon B_1, \ldots, y_m \colon B_m \Rightarrow s : B$$

if, and only if,

$$x_1 \colon A_1, \ldots, x_n \colon A_n \Rightarrow A\text{type} \equiv y_1 \colon B_1, \ldots, y_m \colon B_m \Rightarrow B\text{type}$$

and

$$\vdash x_1 \colon A_1, \ldots, x_n \colon A_n \Rightarrow t = s[y_1 \mid\ x_1, \ldots, y_m \mid\ x_m] : A$$

We define $C(U)$ as the homomorphic image of $R(U)$ induced by this equivalence relation. This means that $C(U)$ is the category of equivalence classes of contexts and of equivalence classes of realizations. Two contexts $x_1 \colon A_1, \ldots, x_n \colon A_n$ and $y_1 \colon B_1, \ldots, y_m \colon B_m$ are equivalent if, and only if,

$$x_1 \colon A_1, \ldots, x_{n-1} \colon A_{n-1} \Rightarrow A_n\text{type} \equiv y_1 \colon B_1, \ldots, y_{m-1} \colon B_{m-1} \Rightarrow B_m\text{type}$$

and two realizations

$$\langle t_1, \ldots, t_m \rangle \colon \langle x_1 \colon A_1, \ldots, x_n \colon A_n \rangle \to \langle y_1 \colon B_1, \ldots, y_m \colon B_m \rangle$$
$$\langle s_1, \ldots, s_m \rangle \colon \langle x'_1 \colon A'_1, \ldots, x'_n \colon A'_n \rangle \to \langle y'_1 \colon B'_1, \ldots, y'_m \colon B'_m \rangle$$

are equivalent if, and only if, for each j, $1 \leq j \leq n$,

$$x_1 \colon A_1, \ldots, x_n \colon A_n \Rightarrow t_j \colon B_j[y_1 \mid\ t_1, \ldots, y_{j-1} \mid\ t_{j-1}] \equiv$$
$$x'_1 \colon A'_1, \ldots, x'_n \colon A'_n \Rightarrow s_j \colon B'_j[y'_1 \mid\ s_1, \ldots, y_{j-1} \mid\ s_{j-1}]$$

It is easy to see that $C(U)$ is itself a contextual category.

Definition 2.12. *if $\mathcal{C}$ and $\mathcal{C}'$ are contextual categories, then a* ***contextual functor*** $F \colon \mathcal{C} \to \mathcal{C}'$ *is a functor $F \colon \mathcal{C} \to \mathcal{C}'$ such that:*

1. $F(1) = 1$ *and if $A \lhd B$ in $\mathcal{C}$, then $F(A) \lhd F(B)$ in $\mathcal{C}'$;*
2. *for all object A of $\mathcal{C}$, $F(p(A)) = p(F(A))$;*
3. *for all f and B such that f^*B is defined in $\mathcal{C}$,*

$$F(f^*B) = F(f)^*F(B) \quad \text{and} \quad F(q(f,B)) = q(F(f), F(B))$$

Con is the category of contextual categories and contextual functors.

2.6 Properties of Contextual Categories

In this section we will state some properties about contextual categories that will be used later.

The first useful result is about the projections, although we defined projections on pairs $A \lhd B$, we might extend the notion to pairs $A \leq B$ defined as

$$p(B, A) = p(X_1) \circ \ldots, \circ p(X_n) \circ B$$

where $X_1, \ldots, X_n$ is the unique sequence of objects in $\mathcal{C}$ such that $A \lhd X_1 \lhd \cdots \lhd X_n \lhd B$ in $\mathcal{C}$. Notation: $B \twoheadrightarrow A$.

Let $\mathcal{C}$ be a contextual category and A an arbitrary object of $\mathcal{C}$. Then we define the **relativization** of $\mathcal{C}$ to A, notes as $\mathcal{C}_A$, as the contextual category where:

1. the objects of $\mathcal{C}_A$ are the objects B of $\mathcal{C}$ such that $A \leq B$.
2. for objects B, C in $\mathcal{C}_A$, a morphism from B to C over A is a morphism, i.e. a morphism from B to C in $\mathcal{A}$, is a morphism $f\colon B \to C$ in $\mathcal{C}$ such that $p(C, A) \circ f = p(B, A)$. Composition of morphisms is inherited from $\mathcal{C}$.
3. $\text{father}_A(B) = \text{father}(B)$ if $A < B$, and $\text{father}_A(A) = A$.
4. for objects $B > A$, $p_A(B) = p(B)\colon B \to \text{father}(B)$.
5. for B, C, D objects of $\mathcal{C}_A$ with $\text{father}_A(D) = C$ and morphisms $f\colon B \to C$ we put
$$f^{*_A} D = f^* D \quad \text{and} \quad q_A(f, D) = q(f, D)$$

Now we will see that any morphism $f\colon B \to A$ in $\mathcal{C}$ induces a pullback functor $f^*\colon \mathcal{C}_A \to \mathcal{C}_B$ which preserves contextual structure.

Lemma 2.13. *let $\mathcal{C}$ be a contextual category, and $f\colon B \to A$ a morphism in $\mathcal{C}$, then for any object C with $C \geq A$ we define an object $f^*C \geq B$ and a morphism $q(f, C)\colon f^*C \to C$ by induction over level$C$ − level$A$:*

1. *if $C = A$,*
$$f^*C = C \quad \textit{and} \quad q(f, C) = id_C$$
2. *if $C > A$,*
$$f^*C = q(f, \textit{father}(C))^*C \quad \textit{and} \quad q(f, C) = q(q(f, \textit{father}(C)), C)$$

then if $C \geq A$, the pair $(p(f^*C, B), q(f, C))$ is a pullback cone over $(f, p(C, A))$.

Theorem 2.14. *Let $\mathcal{C}$ be a contextual category and $f\colon B \to A$ a morphism in $\mathcal{C}$, by mapping an object C of $\mathcal{C}_A$ to f^*C in $\mathcal{C}_B$, and mapping $g\colon C \to D$ over A to $f^*g\colon f^*C \to f^*D$ over B determined by*

$$p(f^*D, B) \circ f^*g = p(f^*C, B)$$
$$q(f, D) \circ f^*g = q(f, C)$$

*we obtain the contextual functor $f^*\colon \mathcal{C}_A \to \mathcal{C}_B$ called **reindexing along** f.*

Theorem 2.15. *Let $\mathcal{C}$ be a contextual category and $f\colon B \to A$, $g\colon C \to B$ be morphisms in $\mathcal{C}$. Then,*

$$g^* \circ f^* = (f \circ g)^* \quad \textit{and} \quad q(f \circ g, D) = q(f, D) \circ q(g, f^*D)$$

for objects $D \geq A$.

Definition 2.16. *Let $\mathcal{C}$ be a contextual category, A and B be objects of $\mathcal{C}$ such that $A \lhd B$, the set of **sections** of B is defined as*

$$\mathit{Sections}(B) := \{s\colon A \to B \mid \ p(B) \circ s = id_A\}$$

If $\mathrm{level}(B) = 1$, then

$$\mathrm{Sections}(B) = \{s \mid \ s\colon 1 \to B\}$$

as any morphism $s\colon 1 \to B$ satisfies $p(B) \circ s = \mathrm{id}_1$ as 1 is a terminal object. In this case, the notion of section matches the notion of *global element* of B.

2.7 Relevance of the context category

To see why this context category is relevant, we need to define interpretation fo a theory U on a contextual category, and see that the interpretation into $C(U)$ is the initial object of the category of interpretations of U.

Given a contextual category $\mathcal{C}$ and a theory U, an **interpretation** of U in $\mathcal{C}$ is a mapping $\mathcal{M}$ from the T- and $\in$-sequents of U to the $\mathcal{C}$ such that

1. if R is a derived T-sequent $\dfrac{x_1\colon A_1, \ldots, x_n\colon A_n}{A \text{ is a type}}$ and we define sequents $R_i \equiv \dfrac{x_1\colon A_1, \ldots, x_{i-1}\colon A_{i-1}}{A_i \text{ is a type}}$ for $1 \leq i \leq n$, then $\mathcal{M}[R_1], \ldots, \mathcal{M}[R_n], \mathcal{M}[R]$ are objects of $\mathcal{C}$ such that

   $$1 \lhd \mathcal{M}[R_1] \lhd \cdots \lhd \mathcal{M}[R_n] \lhd \mathcal{M}[R]$$

   in $\mathcal{C}$.

2. if R_t is a derived $\in$-sequent $\dfrac{x_1 \colon A_1, \ldots, x_n \colon A_n}{t \colon A}$ then $\mathcal{M}[R_t]$ is section of $\mathcal{M}[R]$.

3. if $\Gamma \Rightarrow A = A'$ is a derived T =sequent, then if

$$R \equiv \Gamma \Rightarrow A\text{type} \quad \text{and} \quad R' \equiv \Gamma \Rightarrow A'\text{type}$$

then $\mathcal{M}[R] = \mathcal{M}[R']$.

4. if $\Gamma \Rightarrow t = t' \colon A$ is a derived $\in$=sequent, then if

$$R_t \equiv \Gamma \Rightarrow t \colon A, \quad R_{t'} \equiv \Gamma \Rightarrow t' \colon A, \quad R \equiv \Gamma \Rightarrow A\text{type}$$

then, both $\mathcal{M}[R_t]$ and $\mathcal{M}[R_{t'}]$ are sections of $\mathcal{R}$ and $\mathcal{M}[R_t] = \mathcal{M}[R_{t'}]$

There is a canonical interpretation $\mathcal{M}$ of U in $\mathcal{C}(U)$, defined as follows:

1. if $R \equiv x_1 \colon A_1, \ldots, x_n \colon A_n \Rightarrow A\text{type}$ is a derived T-judgement, then we define

$$\mathcal{M}[R] = [x_1 \colon A_1, \ldots, x_n \colon A_n, x_{n+1} \colon A]$$

2. if $R_t \equiv x_1 \colon A_1, \ldots, x_n \colon A_n \Rightarrow t \colon A$ is a derived $\in$-judgement, then we define

$$\mathcal{M}[R_t] = [\langle x_1, \ldots, x_n, t \rangle]$$

a section of $\mathcal{M}[R]$.

Now we shall define the concept of homomorphism between interpretations of a theory U. Let $\mathcal{C}, \mathcal{C}'$ be two contextual categories, and let $\mathcal{M}_\mathcal{C}$ and $\mathcal{M}_{\mathcal{C}'}$ be two U-interpretations in $\mathcal{C}$ and $\mathcal{C}'$ respectively. Then, an interpretation homomorphism is a contextual functor $F \colon \mathcal{C} \to \mathcal{C}'$ such that:

1. if R is a T-rule, then $F\mathcal{M}_\mathcal{C}[R] = \mathcal{M}_{\mathcal{C}'}[R]$
2. if R_t is a $\in$-rule, then $F\mathcal{M}_\mathcal{C}[R_t] = \mathcal{M}_{\mathcal{C}'}[R_t]$

So we can define the category $\mathbf{Con}_U$ as the category with pairs of contextual categories and U-interpretations and their Homomorphisms. It is easy to see that the pair $(C(U), \mathcal{M})$ defined earlier is an initial object of this category. This means that all the information of U captured in any interpretation, is also captured by this initial interpretation.

Chapter 3

Calculus of Constructions

3.1 Syntactic Formulation

In this section, we will formally define the syntax of Calculus of Constructions. We will use the approach of Thomas Streicher in [4] as the original definition in [3] is characterized by its syntax overloading, making it much harder to follow. They just use $[x : A]B$ to present product types, functional abstraction and universal quantification. Also, properties are presented as terms of type $Prop$ and as types themselves. To solve the latter, we will be using a type constructor $Proof$ that given a proposition $p : Prop$ it will give $Proof(p)$ the type of proofs of proposition p.

First, we will define the class of **pre-well-formed type expressions**, ranged over by capital letters, and the class of **pre-well-formed object expressions**, ranged over by lower case letters, by mutual recursion in a BMF-like way:

$E ::=$	$(\Pi x : E)\, E$	product of types
	$\mid$ Prop	type of propositions
	$\mid$ Proof(t)	type of proofs of proposition t

$t ::=$	x	variable
	$\mid$ $(\lambda x\colon E)t$	functional abstraction
	$\mid$ App$([x\colon E]E, t, t)$	function application
	$\mid$ $(\forall x\colon E)t$	universal quantification

We say a **pre-context** is a syntactical expression of the form $x_1 : A_1, \ldots, x_n : A_n$, where $x_1, \ldots, x_n$ are pairwise syntactically different variables. Contexts are ranged over by capital greek letters. We define $\mathrm{Var}(\Gamma)$ as the set of variables declared in Γ. We will not distinguish expressions which are equivalent up to renaming of

variables (α-conversion), to do this we will be using *deBruijn indices* [1]. We define a **pre-judgment** to be a syntactic expression of one of the following forms:

$A\text{type}$	$A$is a type
$A = B$	A and B are equal types
$t : A$	t is an object of type A
$t = s : A$	t and s are equal objects of type A

we will use the letter $\mathcal{J}$ to refer to pre-judgments. We define a **pre-sequent** as a syntactic expression of the following form

Γok	Γ is a context
$\Gamma = \Delta$	Γ and Δ are equal contexts
$\Gamma \Rightarrow \mathcal{J}$	the judgement $\mathcal{J}$ holds w.r.t. Γ

Finally we define the set of **sequents**, i.e. valid pre-sequents, to be generated by the following set of rules. Sequents are ranged over by capital italic letters. This set of rules is more extent that the original in order to make explicit rules for handling contexts and to add more rules stating equality of types and objects.

When presenting the axiomatization of Calculus of Constructions we shall use the following notation:

$$\frac{S_1}{\frac{}{S_2}} \text{ abbreviates } \frac{S_1}{S_2} \text{ and } \frac{S_2}{S_1}$$

Context Formation Rules:

$$\textsc{Empty}\ \frac{}{\text{ok}}$$

$$\textsc{Cont-Intro}\ \frac{\Gamma \Rightarrow A\text{type}}{\Gamma, x\colon A\text{ok}}, \text{ if } x \notin \text{Var}(\Gamma)$$

$$\textsc{Cont-Thin}\ \frac{\Gamma, \Delta\text{ok} \quad \Gamma \Rightarrow A\text{type}}{\Gamma, x\colon A, \Delta\text{ok}}, \text{ if } x \notin \text{Var}(\Gamma) \cup \text{Var}(\Delta)$$

$$\textsc{Cont-Sub}\ \frac{\Gamma, x\colon A, \Delta\text{ok} \quad \Gamma \Rightarrow t\colon A}{\Gamma, \Delta[x \mid\ t]\text{ok}}$$

Context Equality Rules

$$\textsc{Crefl}\ \frac{\Gamma = \Gamma}{\frac{}{\Gamma\text{ok}}}$$

$$\textsc{Csym}\ \frac{\Gamma = \Delta}{\Delta = \Gamma}$$

[1] *deBruijn indices* are a method to refer to declarations not by name (variables) but by numbers which tell how far one has to go back into the current context of an expression to find the declaration of the thing it refers to.

$$\textsc{Ctrans}\ \frac{\Gamma = \Delta \quad \Delta = \Theta}{\Gamma = \Theta}$$

$$\textsc{Cequ}\ \frac{\Gamma, x\colon A, \Delta\text{ok} \quad \Gamma \Rightarrow A = B}{\Gamma, x\colon A, \Delta = \Gamma, x : B, \Delta}$$

$$\textsc{Creflect1}\ \frac{\Gamma, x\colon A = \Delta, x\colon B}{\Gamma = \Delta}$$

$$\textsc{Creflect2}\ \frac{\Gamma, x\colon A = \Delta, x\colon B}{\Gamma \Rightarrow A = B_1}$$

Structural Rules for Judgement Formation

$$\textsc{Ass}\ \frac{\Gamma, x\colon A, \Delta\text{ok}}{\Gamma, x\colon A, \Delta \Rightarrow x\colon A}$$

$$\textsc{Reflection1}\ \frac{\Gamma \Rightarrow \mathcal{J}}{\Gamma\text{ok}}$$

$$\textsc{Reflection2}\ \frac{\Gamma \Rightarrow t : A}{\Gamma \Rightarrow A\text{type}}$$

$$\textsc{Thin}\ \frac{\Gamma, \Delta \Rightarrow \mathcal{J} \quad \Gamma \Rightarrow A\text{type}}{\Gamma, x\colon A, \Delta \Rightarrow \mathcal{J}}, \text{ if } x \notin \mathrm{Var}(\Gamma) \cup \mathrm{Var}(\Delta)$$

$$\textsc{Sub}\ \frac{\Gamma, x\colon A, \Delta \Rightarrow \mathcal{J} \quad \Gamma \Rightarrow t : A}{\Gamma, \Delta[x \mid\ t] \Rightarrow \mathcal{J}[x \mid\ t]}$$

Equality Rules:

$$\textsc{Equ}\ \frac{\Gamma = \Delta \quad \Gamma \Rightarrow \mathcal{J}}{\Delta \Rightarrow \mathcal{J}}$$

$$\textsc{Refl-Typ}\ \frac{\Gamma \Rightarrow A = A}{\Gamma \Rightarrow A\text{type}}$$

$$\textsc{Refl-Obj}\ \frac{\Gamma \Rightarrow t = t : A}{\Gamma \Rightarrow t : A}$$

$$\textsc{Symm-Typ}\ \frac{\Gamma \Rightarrow A = B}{\Gamma \Rightarrow B = A}$$

$$\textsc{Symm-Obj}\ \frac{\Gamma \Rightarrow t = s : A}{\Gamma \Rightarrow s = t : A}$$

$$\textsc{Trans-Typ}\ \frac{\Gamma \Rightarrow A = B \quad \Gamma \Rightarrow B = C}{\Gamma \Rightarrow A = C}$$

$$\textsc{Trans-Obj}\ \frac{\Gamma \Rightarrow t_1 = t_2 : A \quad \Gamma \Rightarrow t_2 = t_3 : A}{\Gamma \Rightarrow t_1 = t_3 : A}$$

$$\textsc{Repl1}\ \frac{\Gamma \Rightarrow t = s : A \quad \Gamma, x\colon A, \Delta\text{ok}}{\Gamma, \Delta[x \mid\ t] = \Gamma, \Delta[x \mid\ t]}$$

$$\textsc{Repl2}\ \frac{\Gamma \Rightarrow t = s : A \quad \Gamma, x\colon A, \Gamma \Rightarrow B\text{type}}{\Gamma, \Delta[x \mid\ t] \Rightarrow B[x \mid\ t] = B[x \mid\ s]}$$

$$\textsc{Repl3}\ \frac{\Gamma \Rightarrow t_1 = t_2 : A \quad \Gamma, x\colon A, \Delta \Rightarrow s : B}{\Gamma, \Delta[x \mid\ t_1] \Rightarrow s[x \mid\ t_1] = s[x \mid\ t_2] : B[x \mid\ t_1]}$$

$$\textsc{Conv1}\ \frac{\Gamma \Rightarrow A = B \quad \Gamma \Rightarrow t : A}{\Gamma \Rightarrow t : B}$$

$$\textsc{Conv2}\ \frac{\Gamma \Rightarrow A = B \quad \Gamma \Rightarrow t = s : A}{\Gamma \Rightarrow t = s : B}$$

Rules for Products of types

$$\prod\textsc{-Form}\ \frac{\Gamma, x\colon A \Rightarrow B\text{type}}{\Gamma \Rightarrow (\prod x\colon A)B\text{type}}$$

$$\prod\textsc{-Equ}\ \frac{\Gamma \Rightarrow A_1 = A_2 \quad \Gamma, x\colon A_1 \Rightarrow B_1 = B_2}{\Gamma \Rightarrow (\prod x\colon A_1)B_1 = (\prod x\colon A_2)B_2}$$

$$\prod\textsc{-Intro}\ \frac{\Gamma, x\colon A \Rightarrow t : B}{\Gamma \Rightarrow (\lambda x\colon A)t : (\prod x\colon A)B}$$

$$\xi\textsc{-Rule}\ \frac{\Gamma, x\colon A_1 \Rightarrow t_1 = t_2 : B \quad \Gamma \Rightarrow A_1 = A_2}{\Gamma \Rightarrow (\lambda x\colon A_1)t_1 = (\lambda x\colon A_2)t_2\colon (\prod x\colon A_1)B}$$

$$\prod\textsc{-Elim}\ \frac{\Gamma \Rightarrow t\colon (\prod x\colon A)B \quad \Gamma \Rightarrow s\colon A \quad \Gamma, x\colon A \Rightarrow B\text{type}}{\Gamma \Rightarrow \mathrm{App}([x\colon A]B, t, s)\colon B[x \mid\ s]}$$

$$\prod\textsc{-Elim-Equ}\ \frac{\Gamma \Rightarrow A_1 = A_2 \quad \Gamma \Rightarrow t_1 = t_2\colon (\prod x\colon A_1)B_1 \quad \Gamma, x\colon A_1 \Rightarrow B_1 = B_2 \quad \Gamma \Rightarrow s_1 = s_2\colon A_1}{\Gamma \Rightarrow \mathrm{App}([x\colon A_1]B_1, t_1, s_1) = \mathrm{App}([x\colon A_2]B_2, t_2, s_2)\colon B_1[x \mid\ s_1]}$$

$$\beta\textsc{-Rule}\ \frac{\Gamma, x\colon A \Rightarrow B\text{type} \quad \Gamma, x\colon A \Rightarrow t\colon B \quad \Gamma \Rightarrow s\colon A}{\Gamma \Rightarrow \mathrm{App}([x\colon A]B, (\lambda x : A)t, s) = t[x \mid\ s]\colon B[x \mid\ s]}$$

$$\eta\textsc{-rule}\ \frac{\Gamma, x\colon A \Rightarrow B\text{type} \quad \Gamma \Rightarrow t\colon (\prod x\colon A)B}{\Gamma \Rightarrow (\lambda x\colon A)\mathrm{App}([x\colon A]B, t, x) = t\colon (\prod x\colon A)B}$$

Rules for Propositions

$$\textsc{Prop}\ \frac{\Gamma\text{ok}}{\Gamma \Rightarrow \mathrm{Proptype}}$$

$$\textsc{Prop-Type}\ \frac{\Gamma \Rightarrow p\colon \mathrm{Prop}}{\Gamma \Rightarrow \mathrm{Proof}(p)\mathrm{type}}$$

$$\textsc{Prop-Equ}\ \frac{\Gamma \Rightarrow p_1 = p_2\colon \mathrm{Prop}}{\Gamma \Rightarrow \mathrm{Proof}(p_1) = \mathrm{Proof}(p_2)}$$

$$\forall\text{-Intro}\ \frac{\Gamma, x\colon A \Rightarrow p\colon \mathrm{Prop}}{\Gamma \Rightarrow (\forall x\colon A)p\colon \mathrm{Prop}}$$

$$\forall\text{-Equ}\ \frac{\Gamma, x\colon A_1 \Rightarrow p_1 = p_2\colon \mathrm{Prop} \quad \Gamma \Rightarrow A_1 = A_2}{\Gamma \Rightarrow (\forall x\colon A_1)p_1 = (\forall x\colon A_2)p_2\colon \mathrm{Prop}}$$

$$\forall\text{-Elim}\ \frac{\Gamma, x\colon A \Rightarrow p \in \mathrm{Prop}}{\Gamma \Rightarrow \mathrm{Proof}((\forall x\colon A)p) = (\prod x\colon A)\mathrm{Proof}(p)}$$

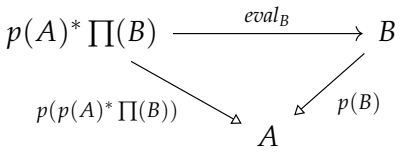
We will write $\vdash \mathcal{J}$ to say that $\mathcal{J}$ is a derivable judgement.

3.2 Doctrines of Constructions

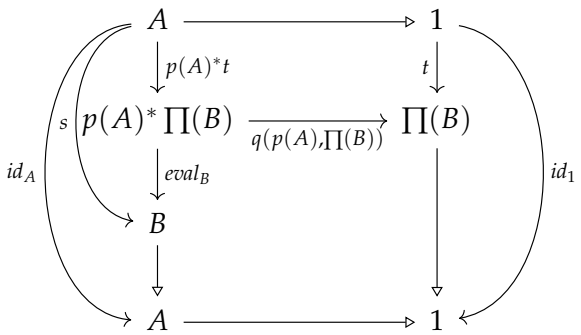
We will leverage the concept of context categories introduced in the earlier chapter to serve as a basic structure for handling dependent types, and add more conditions so that it can handle products of families of types and the type of properties, in order to develop a categorical structure that captures the interpretation of the Calculus of constructions.

Definition 3.1. *A **contextual category with products of families of types** is a category $\mathcal{C}$ together with two mappings $\prod$ and eval defined over the collection of objects B of $\mathcal{C}$ with $level(B) \geq 2$, such that:*

1. *whenever $1 \lhd A \lhd B$, then $1 \lhd \prod(B)$ and $eval_B\colon p(A)^* \prod(B) \to B$ is a morphism over A*



such that for all $s \in Sections(B)$ there is a unique section $t \in Sections(\prod(B))$ with $eval_B \circ p(A)^* t = s$. This unique section t is called $curry(s)$.



Observation 3.2. *as any section t of $\prod(B)$ gives a section $eval_B \circ p(A)^* t$ of B, hence we have a canonical 1-1-correspondence of Sections(B) and Sections$(\prod(B))$.*

2. *and for any object J and any morphism $f : J \to 1$ the following conditions are satisfied*
$$f^*(\prod(B)) = \prod(f^*B) \quad \textit{and} \quad f^*(eval_B) = eval_{f^*B}$$

It follows from the definition that substitution preserves functional abstraction or currying:

Lemma 3.3. *Let $\mathcal{C}$ be a contextual category with products of families of types. Let $1 \lhd A \lhd B$ be objects and $f\colon J \to 1$ a morphism in $\mathcal{C}$. If t is a section of B, then*
$$f^*(curry(B)) = curry(f^*B)$$

Notice that in any contextual category with products of families of types, the following equations hold:
$$\mathrm{eval}_B \circ p(A)^* \mathrm{curry}(s) = s \qquad (\beta)$$
$$\mathrm{curry}(\mathrm{eval}_B \circ p(A)^* t) = t \qquad (\eta)$$

The first equation corresponds to the β-rule of typed λ-calculi and the second equation corresponds to the η-rule of typed λ-calculi. Contextual Categories with products correspond to λ-calculus with dependent types.

Now we will add the type of propositions. To do it, we shall find objects in our category whose sections corresponds, i.e. its terms, correspond to objects.

Definition 3.4. *Let $\mathcal{C}$ be a context category, an **enumeration of types of** $\mathcal{C}$ is an object T of $\mathcal{C}$ with level$(T) = 2$.*

*An enumeration of types T of $\mathcal{C}$ is called a **collection of types** of $\mathcal{C}$ if, and only if, for all objects A and morphisms $f, g\colon A \to father(T)$*
$$f^*T = g^*T \quad \Longrightarrow \quad f = g$$

it is the object father(T) that we are interested in.

Definition 3.5. *a **doctrine of constructions** is a contextual category with products of families of types together with a collection of types T such that for all $A, B \in Ob\mathcal{C}$ with $A \lhd B$ and $f\colon B \to Prop$, where $Prop = fatherT$, there exists a necessarily unique morphism $g\colon A \to Prop$ also denoted by $\forall(f)$ such that*
$$g^*T = \forall(f)^*T = \prod(f^*T)$$

Now we will see that doctrines of constructions correspond to models of Calculus of Constructions.

3.3 Interpretation of Calculus of Constructions

In this section we will define the interpretation of Calculus of Constructions in arbitrary doctrines of constructions.

To do this, fixed one doctrine of constructions, an **interpretation function** $\mathcal{M}$ should be a mapping such that:

1. for every pre-context Γ of length n, if $\mathcal{M}[\Gamma]$ is defined, then $\mathcal{M}[\Gamma]$ is an object of level n;
2. for every pre-context of length n and type expression A, if $\mathcal{M}[\Gamma; A]$ is defined then $\mathcal{M}[\Gamma; A]$ is an object of level $n+1$ and $\mathcal{M}[\Gamma] \lhd \mathcal{M}[\Gamma; A]$;
3. for every pre-context Γ of length n and object expression t, if $\mathcal{M}[\Gamma; t]$ is defined then it is a morphism $s : \mathcal{M}[\Gamma] \to A$ where A is an object of level $n+1$ such that
$$\mathcal{M}[\Gamma] \lhd A \quad \text{and} \quad p(A) \circ s = \mathrm{id}_{\mathcal{M}[\Gamma]}$$

We want to build this function by induction, we need an auxiliary definition:

Definition 3.6. *by structural induction of expressions we define a function* ***depth*** *mapping pre-well-formed expressions and pre-contexts to ω:*

$$depth[x] = 1$$

$$depth[Prop] = 1$$

$$depth[Proof(A)] = depth[A] + 1$$

$$depth[(\prod x\colon A)B] = depth[A] + depth[B] + 1$$

$$depth[(\lambda x\colon A)t] = depth[A] + depth[t] + 1$$

$$depth[(\forall x\colon A)t] = depth[A] + depth[t] + 1$$

$$depth[App([x\colon A]B, t, s)] = depth[A] + depth[B] + depth[t] + depth[s] + 1$$

$$For\ \Gamma \equiv x_1 : A_1, \ldots, x_n : A_n, \quad depth[\Gamma] = \sum_{i=1}^{n} depth[A_i]$$

We will define $\mathcal{M}$ by induction on the level of its arguments where

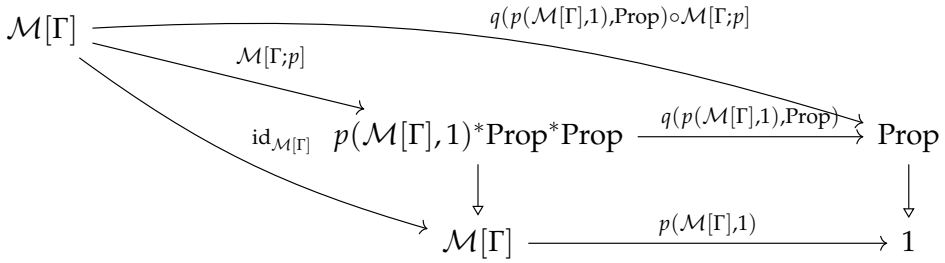
$$\mathrm{level}[\Gamma] = \mathrm{depth}[\Gamma],$$

$$\mathrm{level}[\Gamma; A] = \mathrm{depth}[\Gamma] + \mathrm{depth}[A],$$

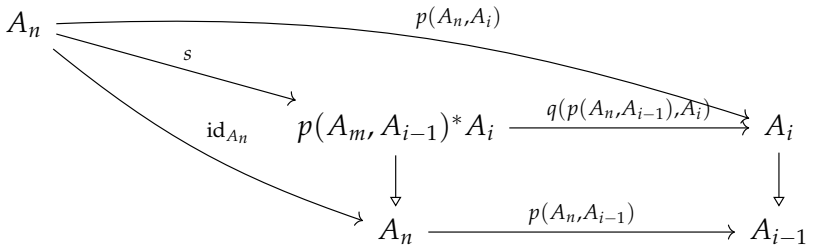
$$\mathrm{level}[\Gamma; t] = \mathrm{depth}[\Gamma] + \mathrm{depth}[t]$$

by the following clauses:

1. $\mathcal{M}[] = 1$
2. $\mathcal{M}[\Gamma, x\colon A] = \mathcal{M}[\Gamma; A]$
3. $\mathcal{M}[\Gamma, (\prod x\colon A)B] = \prod(\mathcal{M}[\Gamma, x\colon A; B])$ if $\mathcal{M}[\Gamma, x\colon A; B]$ is defined
4. $\mathcal{M}[\Gamma, \mathrm{Prop}] = p(\mathcal{M}, 1)^*\mathrm{Prop}$ if $\mathcal{M}[\Gamma]$ is defined
5. $\mathcal{M}[\Gamma; \mathrm{Proof}(p)] = \mathcal{M}[\Gamma; p]^* p(\mathcal{M}[\Gamma], 1)^* T$ if $\mathcal{M}[\Gamma; p]$ is defined and a section of $p(\mathcal{M}[\Gamma], 1)^*\mathrm{Prop}$



6. $\mathcal{M}[\Gamma, x] = s$ provided that $\Gamma \equiv x_1\colon A_1, \ldots, x_n\colon A_n$ and $x_i \equiv x$ and $\mathcal{M}[\Gamma]$ is defined and $\mathcal{M}[\Gamma] = A_n \triangleright \cdots \triangleright A_1 \triangleright A_0 \triangleright 1$ and s as follows



   or in other terms $\mathcal{M}[\Gamma; x_i] = p(A_n, A_i)^*\Delta(A_i)$ where $\Delta(A_i)$ is the unique arrow $m : A_i \to p(A_i)^* A_i$ such that

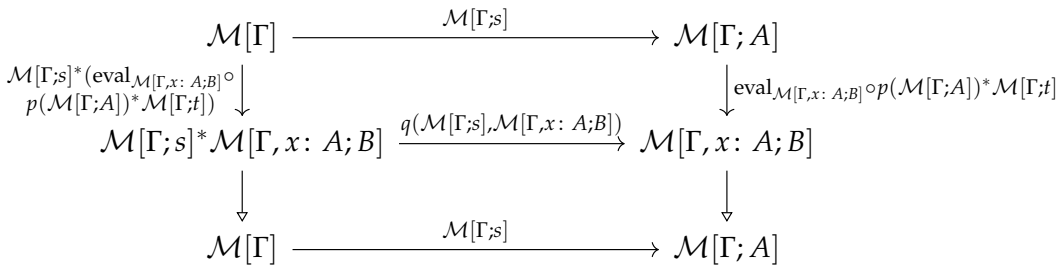
$$p(p(A_i)^* A_i) \circ m = q(p(A_i), A_i) \circ m = id_{A_i}$$

7. $\mathcal{M}[\Gamma; (\lambda x\colon A)t] = \mathrm{curry}(\mathcal{M}[\Gamma, x\colon A; t])$ if $\mathcal{M}[\Gamma, x\colon A; t]$ is defined. By induction hypothesis we have that $\mathcal{M}[\Gamma] \lhd \mathcal{M}[\Gamma; A] \lhd \mathrm{cod}(\mathcal{M}[\gamma, x\colon A; t])$ and s is a section of $\mathcal{M}[\Gamma, x\colon A; t]$ and therefore we have:

$$\mathrm{eval}_{\mathrm{cod}(\mathcal{M}[\Gamma, x\colon A; t])} \circ p(\mathcal{M}[\Gamma; A])^* \mathcal{M}[\Gamma; (\lambda x\colon A)t] = \mathcal{M}[\Gamma, x\colon A; t]$$

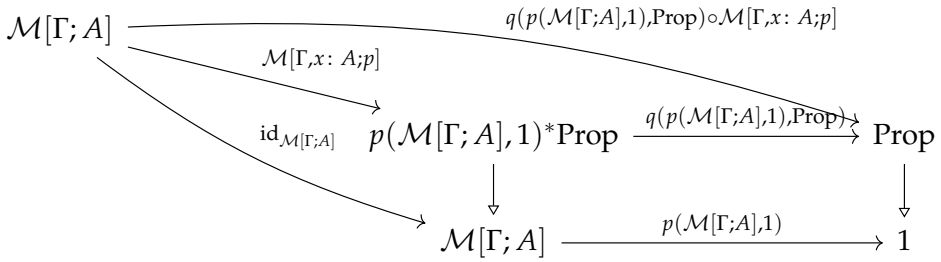
8. $\mathcal{M}[\Gamma; \mathrm{App}([x\colon A]B), t, s] = \mathcal{M}[\Gamma; s]^*(\mathrm{eval}_{\mathcal{M}[\Gamma, x\colon A; B]} \circ p(\mathcal{M}[\Gamma; A])^* \mathcal{M}[\Gamma; t])$ provided that $\mathcal{M}[\Gamma; t]$, $\mathcal{M}[\Gamma; s]$, $\mathcal{M}[\Gamma, x\colon A; B]$ are defined, and $\mathcal{M}[\Gamma; s]$ is a

section of $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; t]$ is a section of $\prod(\mathcal{M}[\Gamma, x\colon A; B])$



9. $\mathcal{M}[\Gamma; (\forall x\colon A)p] = s$ provided $\mathcal{M}[\Gamma, x\colon A; p]$ is a section of $p(\mathcal{M}[\Gamma; A], 1)^*\mathrm{Prop}$ and s is the unique section of $p(\mathcal{M}[\Gamma], 1)$ such that

$$\forall(q(p(\mathcal{M}[\Gamma; A], 1), \mathrm{Prop}) \circ \mathcal{M}[\Gamma, x\colon A; p]) = q(p(\mathcal{M}[\Gamma; A], 1), \mathrm{Prop}) \circ \mathcal{M}[\Gamma, x\colon A; p]$$



Theorem 3.7 (Correctness Theorem)**.** *for any doctrine of constructions the following correctness conditions for $\mathcal{M}$ are satisfied:*

1. *if $\vdash \Gamma ok$ then $\mathcal{M}[\Gamma]$ is defined;*
2. *if $\vdash \Gamma \Rightarrow A type$ then $\mathcal{M}[\Gamma; A]$ is defined;*
3. *if $\vdash \Gamma \Rightarrow t\colon A$ then $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; t]$ are both defined and $\mathcal{M}[\Gamma; t]$ is a section of $\mathcal{M}[\Gamma; A]$;*
4. *if $\vdash \Gamma = \Delta$ then $\mathcal{M}[\Gamma]$ and $\mathcal{M}[\Delta]$ are defined and $\mathcal{M}[\Gamma] = \mathcal{M}[\Delta]$;*
5. *if $\vdash \Gamma \Rightarrow A = B$ then $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; B]$ are both defined and $\mathcal{M}[\Gamma; A] = \mathcal{M}[\Gamma; B]$;*
6. *if $\vdash \Gamma \Rightarrow t = s \in A$ then $\mathcal{M}[\Gamma; A]$, $\mathcal{M}[\Gamma; t]$ and $\mathcal{M}[\Gamma; s]$ are defined, both $\mathcal{M}[\Gamma; t]$ and $\mathcal{M}[\Gamma; s]$ are sections of $\mathcal{M}[\Gamma; A]$ and $\mathcal{M}[\Gamma; t] = \mathcal{M}[\Gamma; s]$.*

Hence, doctrines of constructions are models of Calculus of Constructions.

3.4 The Term Model of Calculus of Constructions

In this section we will construct a doctrine of constructions $\mathcal{C}_I$ as a *term model* of calculus of constructions such that the interpretation in this model interprets provably well-defined contexts, types and objects, and identifies only those contexts, types or objects which are provably equal up to renaming of the variables bound in contexts.

In more technical terms, that means that for the interpretation function $\mathcal{M}$ assigning meaning to calculus of constructions in the doctrine of constructions $\mathcal{C}_I$, the following requirements must be fulfilled:

1. for all pre-contexts Γ and Δ, if $\mathcal{M}[\Gamma] = \mathcal{M}[\Delta]$, then $\vdash \Gamma = \Delta$;
2. for all pre-contexts Γ and Δ and pre-type-expressions A and B, if $\mathcal{M}[\Gamma; A] = \mathcal{M}[\Delta; B]$, then $\vdash \Gamma = \Delta$ and $\vdash \Gamma \Rightarrow A = B$;
3. for all pre-contexts Γ and Δ, pre-type-expressions A and B and pre-object-expressions t and s, if $\mathcal{M}[\Gamma; t] = \mathcal{M}[\Delta; s]$ is a section of $\mathcal{M}[\Gamma; A] = [\Delta; B]$, then $\vdash \Gamma = \Delta$, $\vdash \Gamma \Rightarrow A = B$ and $\vdash \Gamma \Rightarrow t = s : A$.

In order to be able to identify contexts up to α-conversion, we will consider a fixed numerable set of variables: $v_1, \ldots, v_n, \ldots$. We will say that a context $\Gamma \equiv x_1 : A_1, \ldots, x_n : A_n$ is in **α-normal form** iff $x_i \equiv v_i$ for all $1 \leq i \leq n$. Moreover, we will say that a pre-context Γ is admissible iff it is in α-normal form and $\vdash \Gamma$ok.

Now we will define $\mathcal{C}_I$:

The objects of $\mathcal{C}_I$ are the admissible contexts modulo provable equivalence, i.e. that we will consider two contexts Γ and Δ equal iff $\vdash \Gamma = \Delta$. By induction over the structure of derivations, one can see that $\vdash \Gamma = \Delta$ implies that Γ and Δ have the same length as pre-contexts, so we can define the level of an object of $\mathcal{C}_I$ as the length of any of its representatives. We will write $[\Gamma]$ to represent the class containing Γ. If $\Gamma \equiv v_1 : A_1, \ldots, v_n : A_n$ is an admissible context, then we define

$$\text{father}[\Gamma] = [v_1 : A_1, \ldots, v_{n-1} : A_{n-1}]$$

that is well defined according to rule CREFL1

If $\Gamma \equiv v_1 : A_1, \ldots, v_n : A_n$ and $\Delta \equiv v_1 : B_1, \ldots, v_m : B_m$ are admissible contexts, a morphism from $[\Gamma]$ to $[\Delta]$ is given by a m-tuple $\langle t_1, \ldots, t_m \rangle$ of terms such that for i, $1 \leq i \leq m$,

$$\vdash \Gamma \Rightarrow t_i : B_i[v_1 \mid t_1, \ldots, v_{i-1} \mid t_{i-1}]$$

if $\langle s_1, \ldots, s_m \rangle$ is another m-tuple such that for i, $1 \leq i \leq m$

$$\vdash \Gamma \Rightarrow s_i : B_i[v_1 \mid s_1, \ldots, v_{i-1} \mid s_{i-1}]$$

then $\langle t_1, \ldots, t_m \rangle$ and $\langle s_1, \ldots, s_m \rangle$ are considered equal iff for i, $1 \leq i \leq m$

$$\vdash \Gamma \Rightarrow t_i = s_i : B_i[v_1 \mid s_1, \ldots, v_{i-1} \mid s_{i-1}]$$

according to rules CREFL1 and CREFL2 the definition is independent from the choice of representatives. We write $[\langle t_1, \ldots, t_m \rangle] : [\Gamma] \to [\Delta]$ for the class of m-tuples equatable to $\langle t_1, \ldots, t_m \rangle$. If we consider $\Theta \equiv v_1 : C_1, \ldots, v_k : C_k$ is another admissible context and $[\langle s_1, \ldots, s_k \rangle] : [\Delta] \to [\Theta]$, then the composition of $[\langle t_1, \ldots, t_m \rangle]$ with $[\langle s_1, \ldots, s_k \rangle]$ is defined as

$$[\langle s_1, \ldots, s_k \rangle] \circ [\langle t_1, \ldots, t_m \rangle] = [\langle s_1[v_1 \mid t_1, \ldots, v_m \mid t_m], \ldots, s_k[v_1 \mid t_1, \ldots, v_m \mid t_m] \rangle]$$

If $\Gamma \equiv v_1 : A_1, \ldots, v_n : A_n$ is an admissible context with $n \geq 1$ then the *canonical projection* map $p([\Gamma]) : [\Gamma] \to \text{father}[\Gamma]$ is represented by the tuple $\langle v_1, \ldots, v_{n-1} \rangle$. Let $\Gamma \equiv v_1 : A_1, \ldots, v_n : A_n$ and $\Delta \equiv v_1 : B_1, \ldots, v_m : B_m$ admissible contexts and $t = [\langle t_1, \ldots, t_m \rangle] : [\Gamma] \to [\Delta]$ and C a pre-well-formed type expression with $\vdash \Delta \Rightarrow C\text{type}$, i.e. $\Delta, v_{m+1} : C$ is an admissible context, then

$$t^*[\Delta, v_{m+1} : C] = [\Delta, v_{n+1} : C[v_1 \mid t_1, \ldots, v_m \mid t_m]]$$

$$q([t], [\Delta, v_{m+1} : C]) = [\langle t_1, \ldots, t_m, v_{n+1} \rangle] : t^*[\Delta, v_{m+1} : C] \to [\Delta, v_{m+1} : C]$$

If $\Gamma \equiv v_1 : A_1, \ldots, v_n : A_n, v_{n+1} : B, v_{n+2} : C$ is an admissible context we define

$$\prod([\Gamma]) = [\Gamma, v_{n+1} : (\prod v_{n+1} : B)C]$$

$$ev_{[\Gamma]} = [\langle v_1, \ldots, v_n, v_{n+1}, \text{App}([v_{n+1} : B]C, v_{n+2}, v_{n+1}) \rangle]$$

which can be seen to be a morphism from $[\Gamma, v_{n+1} : B, v_{n+2} : (\prod v_{n+1} B)C]$ to $[\Gamma]$ over the object $[v_1 : A_1, \ldots, v_n : A_n]$. The object Prop of C_I is defined as $[v_1 : \text{Prop}]$ and the object T of C_I is defined as the object $[v_1 : \text{Prop}, v_2 : \text{Proof}(v_1)]$. If $\Gamma \equiv v_1 : A_1, \ldots, v_n : A_n, v_{n+1} : B$ and $[\langle p \rangle]$ is a morphism from $[\Gamma]$ to $[v_1 : \text{Prop}]$ then we define

$$\forall([p]) = [\langle (\forall v_{n+1} : B)p \rangle] : [v_1 : A_1, \ldots, v_n : A_n] \to [v_1 : \text{Prop}]$$

It is proven in [4] that C_I is in fact a doctrine of constructions. Also, it is immediate to see that C_I does not interpret or identify more objects or terms that those that are provably defined as equal.

Theorem 3.8 (Completeness Theorem)**.** *Let $\mathcal{M}$ be the interpretation of calculus of construction in C_I, then*

1. *if $\mathcal{M}[\Gamma]$ is defined then $\vdash \Gamma ok$*

2. *if* $\mathcal{M}[\Gamma; A]$ *is defined then* $\vdash \Gamma \Rightarrow A type$

3. *if* $\mathcal{M}[\Gamma; t]$ *is defined then* $\vdash \Gamma \Rightarrow t : A$ *for some pre-type-expression* A

4. *if* $\Gamma \equiv x_1 : A_1, \ldots, x_n : A_n$, $\Delta \equiv y_1 : B_1, \ldots, y_m : B_m$ *and* $\mathcal{M}[\Gamma; A] = \mathcal{M}[\Delta; B]$, *then* $n = m$ *and*

$$\begin{aligned} \vdash v_1 : A_1[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n], \ldots, v_n : A_n[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n] = \\ = v_1 : B_1[y_1 \mid\ v_1, \ldots, y_m \mid\ v_m], \ldots, v_m : B_m[y_1 \mid\ v_1, \ldots, y_m \mid\ v_m] \end{aligned}$$

$$\begin{aligned} \vdash v_1 : A_1[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n], \ldots, v_n : A_n[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n] \\ \Rightarrow A[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n] = B[y_1 \mid\ v_1, \ldots, y_m \mid\ v_m] \end{aligned}$$

5. *if* $\Gamma \equiv x_1 : A_1, \ldots, x_n : A_n$, $\Delta \equiv y_1 : B_1, \ldots, y_m : B_m$ *and* $\mathcal{M}[\Gamma; t] = \mathcal{M}[\Delta, s]$, *then* $n = m$ *and*

$$\begin{aligned} \vdash v_1 : A_1[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n], \ldots, v_n : A_n[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n] = \\ = v_1 : B_1[y_1 \mid\ v_1, \ldots, y_m \mid\ v_m], \ldots, v_m : B_m[y_1 \mid\ v_1, \ldots, y_m \mid\ v_m] \end{aligned}$$

and for some pre-type-expression A

$$\begin{aligned} \vdash v_1 : A_1[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n], \ldots, v_n : A_n[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n] \\ \Rightarrow t[x_1 \mid\ v_1, \ldots, x_n \mid\ v_n] = s[y_1 \mid\ v_1, \ldots, y_m \mid\ v_m] : A \end{aligned}$$

This result says that $\mathcal{C}_I$ is the initial object of the category of doctrines of constructions and structure preserving functors.

Chapter 4

Expansion Algorithm

4.1 Problem Definition

In this section we will introduce a formal definition of our problem, proposed at the beginning of the thesis, and reformulate it to an equivalent problem easier to solve.

Our problem might be formalized in the following manner: given a theory U and a derived rule of U of the form

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$$

we want to find an expression of U such that the rule

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

is a derived rule of U.

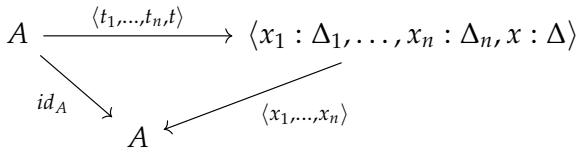
This problem is undecidable, in general, as it follows from the undecidability of the inhabitation in a GAT. This is why we won't try to find an algorithm that either finds a term or says that the type is empty. We aim to find an algorithm that in the case that the type is inhabited, it will find a term in a finite number of steps. In the case that the type is empty, it does not have to give an output.

We can consider this problem from the point of view of the context category $C(U)$, where it is rephrased as: given a context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ in $C(U)$, we want to find a morphism

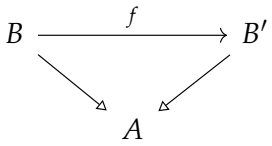
$$\langle x_1, \ldots, x_n, t \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \to \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x : \Delta \rangle$$

in $C(U)$. Then if we name $A = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$, this is equivalent to find a morphism $\langle t_1, \ldots, t_n, t \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \to \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x : \Delta \rangle$ in the

category C_A, as by the following commutative diagram



for each $1 \leq i \leq n$, $x_i = t_i$. This restriction holds for all morphism $f = \langle t_1, \ldots, t_n, t_{n+1}, \ldots, t_{n+m}\rangle : B \to B'$ in C_A, by the commutativity of the diagram



$t_i = x_i$ for all $1 \leq i \leq n$. That's why we will adopt the notation $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A$ to represent $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle$ and $f = \langle t_{n+1}, \ldots, t_{n+m}\rangle_A$ to represent $\langle x_1, \ldots, x_n, t_{n+1}, \ldots, t_{n+m}\rangle$ in C_A.

4.2 Expansion algorithm

In this section we will expose the expansion algorithm, that will be responsible of finding the $\langle t\rangle_A$ in C_A.

First consider G a finite subcategory of C_A such that it contains $\langle\rangle_A$, $\langle x : \Delta\rangle_A$ and $p(\langle x : \Delta\rangle_A) : \langle x : \Delta\rangle_A \to \langle\rangle_A$. The process will consist in the construction of a sequence $\{G_i\}_i$ of finite subcategories of C_A such that $G_0 = G$ and that G_{n+1} is a finite subcategory of C_A computed via the process described below from G_n and where $G_n \subseteq G_{n+1}$ for all $n \geq 0$. We say that an object/morphism of C_A is **constructible by expansion** if there exists a $n \geq 0$ such that the given object/morphism is contained in G_n. The process of constructing G_{n+1} from G_n is consists on applying the following rules to each context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A$ of G_n.

E1 For each sort symbol A with a well-formed introductory rule

$$\frac{z_1 : \Lambda_1, \ldots, z_n : \Lambda_n}{A(z_1, \ldots, z_n) \text{ is a type}}$$

such that G_n contains morphisms

$$\begin{aligned}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_1\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1 \rangle_A \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_2\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_2 : \Lambda_2[t_1|z_1] \rangle_A \\
&\vdots \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_w\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}] \rangle_A
\end{aligned}$$

then we add the context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, y : A(t_1, \ldots, t_w) \rangle_A$ and its projection to G_{n+1}.

E2 add the morphisms

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,y_i\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, y : \Omega_i \rangle_A$$

and their corresponding projections for each $0 \leq i \leq n$.

E3 for every operator symbol F with a well-formed introductory rule

$$\frac{z_1 : \Lambda_1, \ldots, z_n : \Lambda_n}{F(z_1, \ldots, z_n) : \Lambda}$$

if the following morphisms exist

$$\begin{aligned}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_1\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1 \rangle_A \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_2\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_2 : \Lambda_2[t_1|z_1] \rangle_A \\
&\vdots \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_w\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}] \rangle_A
\end{aligned}$$

then we add the morphism

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,F(t_1,\ldots,t_w)\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda[t_1|z_1, \ldots, t_w|z_w] \rangle_A$$

E4 for every morphism

$$\langle t_1, \ldots, t_{m-1} \rangle_A : \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A \to \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1} \rangle_A$$

we construct the corresponding pullback

$$\begin{array}{ccc} \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w, y_m : \Omega_m[t_1|y_1, \ldots, t_{m-1}|y_{m-1}]\rangle_A & \xrightarrow{\langle t_1,\ldots,t_{m-1},y_m\rangle_A} & \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A \\ \downarrow & & \downarrow \\ \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w\rangle_A & \xrightarrow{\langle t_1,\ldots,t_{m-1}\rangle_A} & \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A \end{array}$$

in G_{n+1}.

Once we have applied the expansion rules to all of the contexts, we complete G_{n+1} with the composition morphisms to ensure it is a subcategory.

Theorem 4.1. *Every object and every morphism of C_A is constructible by expansion.*

Proof. We will proof this result by structural induction over the underlying derived rules corresponding to such object (context) or morphism (operator).

Consider a context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A$ of U, which is the representative of an object in $C(U)$. This means, by the definition of context, that its underlying rule

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{\Omega_m \text{ is a type}} \tag{4.1}$$

is a derived rule of U and y_m is a variable different to $x_1, \ldots, x_n, y_1, \ldots, y_{m-1}$. Hence, the premise $\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A$ is a context and by induction it is constructible by expansion, i.e. there exists $n_0 \in \mathbb{N}$ such that $\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A \in G_{n_0}$. As (4.1) is a derived rule, it must have been derived by the only derivation rule that has a T-rule as a conclusion: CF2(a). Hence, there exists a sort symbol S with a well-formed introductory rule $\dfrac{z_1 : \Lambda_1, \ldots, z_w : \Lambda_w}{S(z_1, \ldots, z_w) \text{ is a type}}$ and derived rules

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{t_1 : \Lambda_1},$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{t_2 : \Lambda_2[t_1|z_1]},$$
$$\vdots$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{t_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}]}$$

By induction hypothesis their corresponding morphisms

$$\langle y_1, \ldots, y_{m-1}, t_j\rangle_A : \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A \to \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}, z_j : \Lambda[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]\rangle_A$$

are constructible by expansion, so there exist in G_{n_j} for some $n_j \in \mathbb{N}$.

Take $n = \max\{n_0, n_1, \ldots, n_j\}$. Then, all of the objects, and morphisms mentioned belong to G_n so by the expansion rule E1,

$$\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1} \rangle_A \longleftarrow \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}, y_m : S(t_1, \ldots, t_w) \rangle_A$$

is included in G_{n+1}. And by the equivalence of intended equivalence of denotation

$$\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}, y_m : S(t_1, \ldots, t_w) \rangle_A \equiv \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$$

so $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$ is constructible by expansion.

Now consider an operator

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle t_1, \ldots, t_w \rangle} \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A$$

a representative of a morphism in C_A. By induction hypothesis we know that both $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$ and $\langle z_1 : \Omega_1, \ldots, z_w : \Omega_w \rangle_A$ are constructible by expansion.

Moreover, as $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$ is an operator, for each $1 \leq j \leq w$, the rule

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{t_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]} \tag{4.2}$$

is a derived rule of U. Now we will see that the following morphisms are constructible by expansion:

$$\begin{aligned}
&\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_1 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1 \rangle_A \\
&\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_2 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_2 : \Lambda_2[t_1|z_1] \rangle_A \\
&\qquad\qquad\qquad\qquad \vdots \\
&\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_w \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}] \rangle_A
\end{aligned}$$

To do this, we will use how any of the rules (4.2) are derived. To do it, we must use one of the following derivation rules: T1, CF1 or CF2(b). Hence,

- if derived by T1, we have that both

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{\Delta = \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]}$$

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{t_j : \Delta}$$

are derived rules of U. So by the first rule, due to the equivalence realization of identity of denotation, we know that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Delta \rangle_A \equiv \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$$

hence they are the same object in C_A. And by the second one, applying induction, we have that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Delta \rangle_A$$

is constructible by expansion, so as $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Delta \rangle_A$ and $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$ are the same object in C_A, then

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$$

is constructible by expansion.

- if derived by CF1, we have that

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{\Omega_m \text{ is a type}}$$

is a derived rule of U, where y_m is a variable distinct from all $x_1, \ldots, x_n$, $y_1, \ldots, y_{m-1}$, and that there exists an i, $1 \leq i \leq m$, such that

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{t_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]} \equiv \frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{y_i : \Omega_i}$$

Then, by induction hypothesis, the context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ is constructible by expansion. By applying the expansion rule E2 and an argument of equivalence similar to the one above, we get that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$$

is constructible by expansion.

- if derived by CF2(b), there exists an operator symbol F with a well-formed introductory rule

$$\frac{c_1 : \Gamma_1, \ldots, c_v : \Gamma_v}{F(c_1, \ldots, c_v) : \Gamma}$$

and derived rules

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{s_1 : \Gamma_1}$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{s_2 : \Gamma_2[s_1|c_1]}$$
$$\vdots$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{s_v : \Gamma_v[s_1|c_1, \ldots, s_{v-1}|c_{v-1}]}$$

such that

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{\Gamma[s_1|c_1, \ldots, s_v|c_v] = \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]}$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{t_j = F(s_1, \ldots, s_v) : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]}$$

are derived rules of U. By the equivalence relation of identity of denotation, we have to prove that the morphism

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, F(s_1, \ldots, s_v) \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Gamma[s_1|c_1, \ldots, s_v|c_v] \rangle_A$$

is constructible by expansion. By induction we have that the following morphisms are constructible by expansion:

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, s_1 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, c_1 : \Gamma_1 \rangle_A$$
$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, s_2 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, c_2 : \Gamma_2[s_1|c_1] \rangle_A$$
$$\vdots$$
$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, s_v \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, c_v : \Gamma_v[s_1|c_1, \ldots, s_{v-1}|c_{v-1}] \rangle_A$$

Hence, by applying expansion rule E3, we have that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, F(s_1, \ldots, s_v) \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Gamma[s_1|c_1, \ldots, s_v|c_v] \rangle_A$$

is constructible by expansion as we wanted.

Now we will show tha the morphism

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_1, \ldots, t_w \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A$$

is constructible by expansion. To do this we will prove that given a j, $2 < j \leq n$, if

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_1, \ldots, t_{j-1} \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_{j-1} : \Lambda_{j-1} \rangle_A$$

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$$

are constructible by expansion so is

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,t_1,\ldots,t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_j : \Lambda_j - 1 \rangle_A$$

By expansion rule E4, the following diagram is constructible by expansion:

$$\begin{array}{ccc}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_j|z_j] \rangle_A & \xrightarrow{\langle y_1,\ldots,y_m,t_1,\ldots,t_{j-1},z_j \rangle_A} & \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_j : \Lambda_j \rangle_A \\
\downarrow & & \downarrow \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A & \xrightarrow{\langle y_1,\ldots,y_m,t_1,\ldots,t_{j-1} \rangle_A} & \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_{j-1} : \Lambda_{j-1} \rangle_A
\end{array}$$

and by composition we get that

$$\langle y_1, \ldots, y_m, t_1, \ldots, t_{j-1}, z_j \rangle_A \circ \langle y_1, \ldots, y_m, t_j \rangle_A = \langle y_1, \ldots, y_m, t_1, \ldots, t_j \rangle_A$$

is constructible by expansion. It remains to see that the morphsim

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A \xrightarrow{\langle z_1,\ldots,z_w \rangle_A} \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle$$

is constructible by expansion. That comes directly from the fact that $\langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A$ is a context and the repeated application of the rule E4 as this diagram is constructible by expansion:

$$\begin{array}{ccc}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m z_1 : \Lambda_1, \ldots, z_w : \Omega_w \rangle_A & \xrightarrow{\langle z_1,\ldots,z_w \rangle_A} & \langle z_1 : \Lambda_1, \ldots, z_w : \Omega_w \rangle_A \\
\downarrow & & \downarrow \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m z_1 : \Lambda_1, \ldots, z_{w-1} : \Omega_{w-1} \rangle_A & \xrightarrow{\langle z_1,\ldots,z_{w-1} \rangle_A} & \langle z_1 : \Lambda_1, \ldots, z_{w-1} : \Omega_{w-1} \rangle_A \\
\downarrow & & \downarrow \\
\vdots & & \vdots \\
\downarrow & & \downarrow \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1 \rangle_A & \xrightarrow{\langle z_1 \rangle_A} & \langle z_1 : \Lambda_1 \rangle_A \\
\downarrow & & \downarrow \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle & \xrightarrow{\langle \rangle_A} & \langle \rangle_A
\end{array}$$

□

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